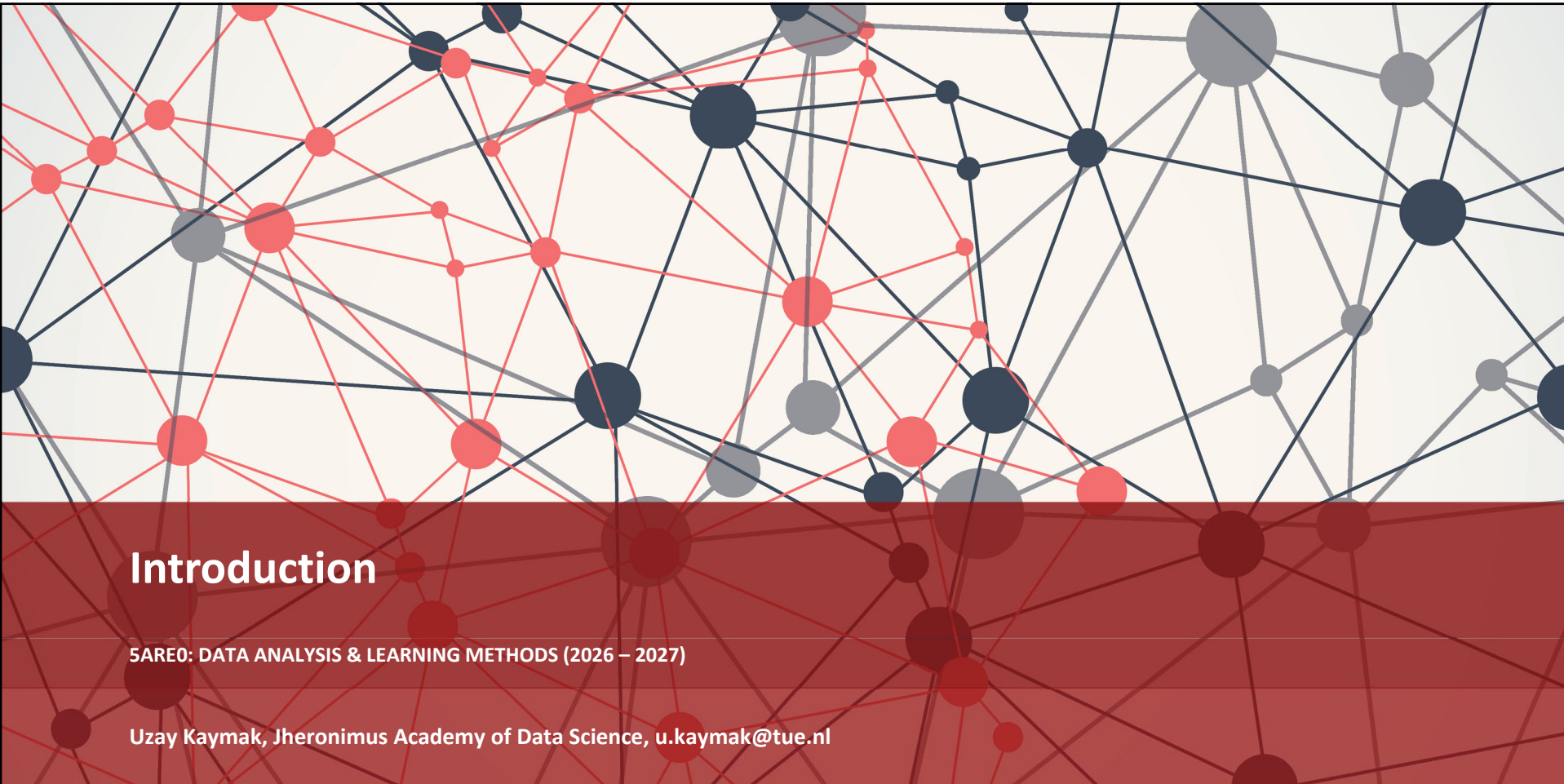




Introduction

5ARE0: DATA ANALYSIS & LEARNING METHODS (2026 – 2027)

Uzay Kaymak, Jheronimus Academy of Data Science, u.kaymak@tue.nl

Master: Artificial Intelligence & Engineering Systems

Outline

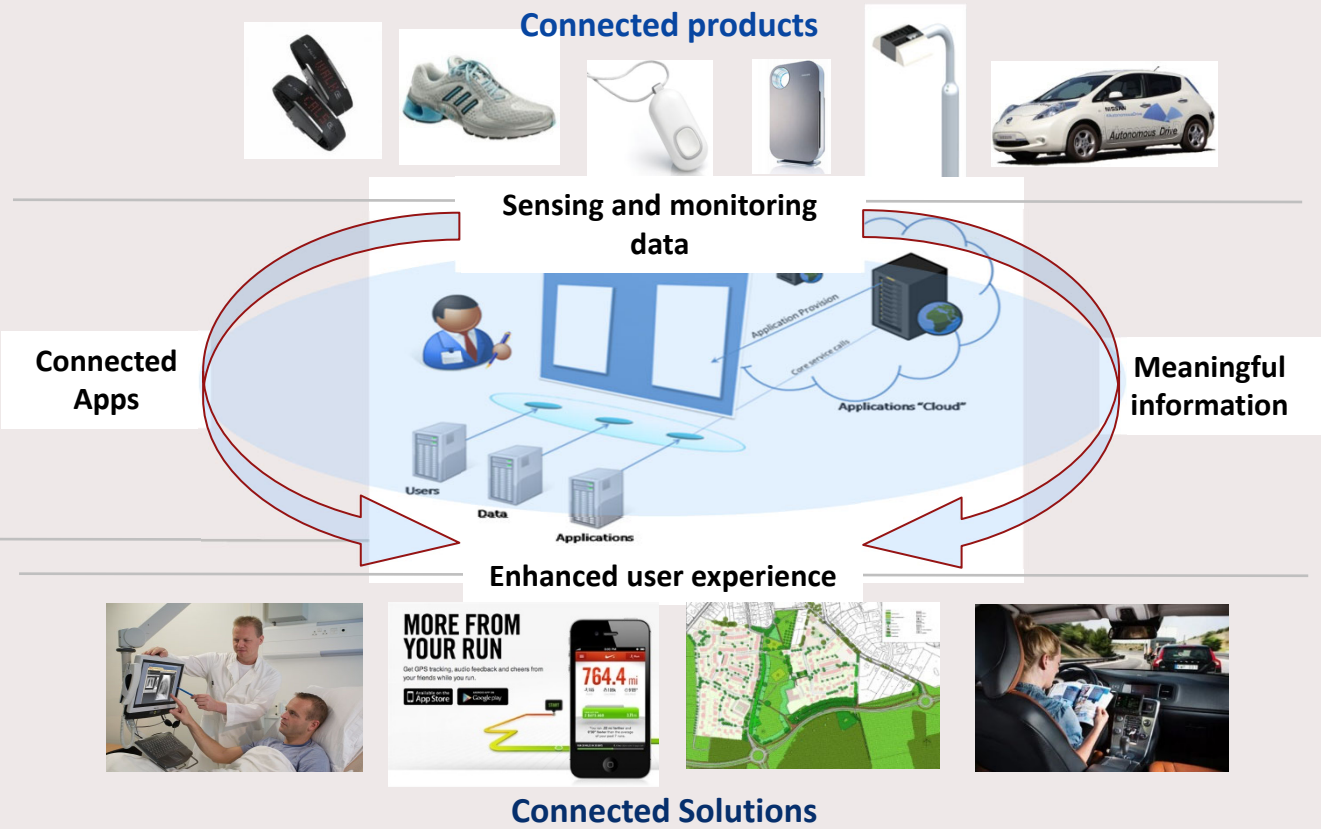
Motivation

Course organization

Artificial intelligence & engineering systems

Preliminaries: modeling with data

The Always-On Society



Everywhere Analytics

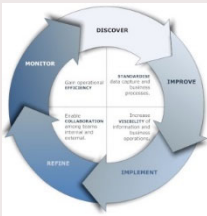
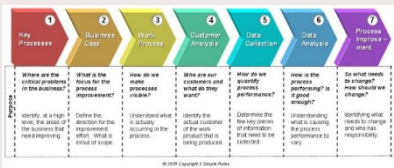
From Deloitte



Real-world examples

Smart business solutions

Process improvement



Monitors and analyses events in an organization and proposes business improvement actions.

Smart power grids



Measures, monitors, and manages energy production, transport, and consumption is heterogeneous distributed grids.

Clinical decision support



Provides instant clinical decision support by correlating information from different part of uncorrelated sources.

AI systems: self-driving cars



Maker: metamorworks | Bronvermelding: Getty Images/iStockphoto

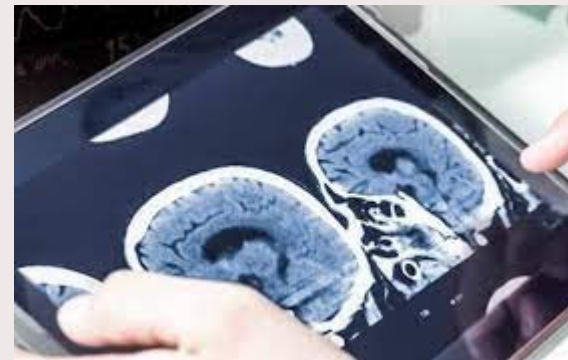


<https://getfello.com/wp-content/uploads/2017/03/Google-self-driving-car-prototype-front-three-quarters1.jpg>

AI Systems: IBM Watson

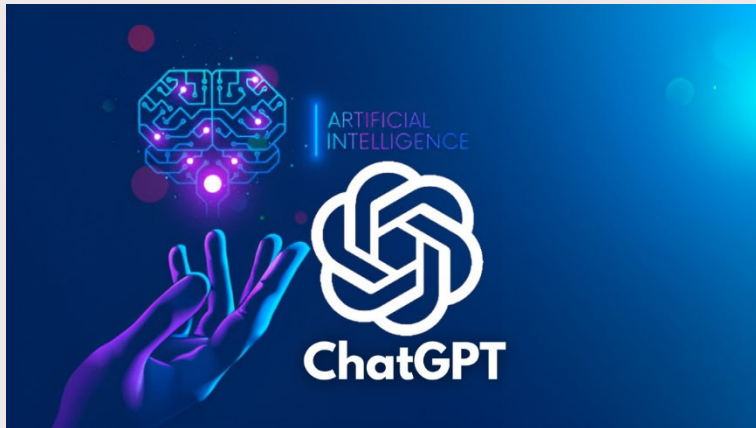


[IBM Watson Health focused on value-based care, physician burnout, personalized medicine | Healthcare IT News](#)



[IBM Watson moet helpen bij uitlezen röntgenfoto's - ICT&health \(icthealth.nl\)](#)

AI Systems: ChatGPT



<https://hawar.no/2023/04/what-is-chatgpt-how-do-you-use-it-how-do-you-benefit-from-it/>



<https://www.datacamp.com/blog/a-chat-with-chatgpt-on-the-method-behind-the-bot>

AI Systems: robot football



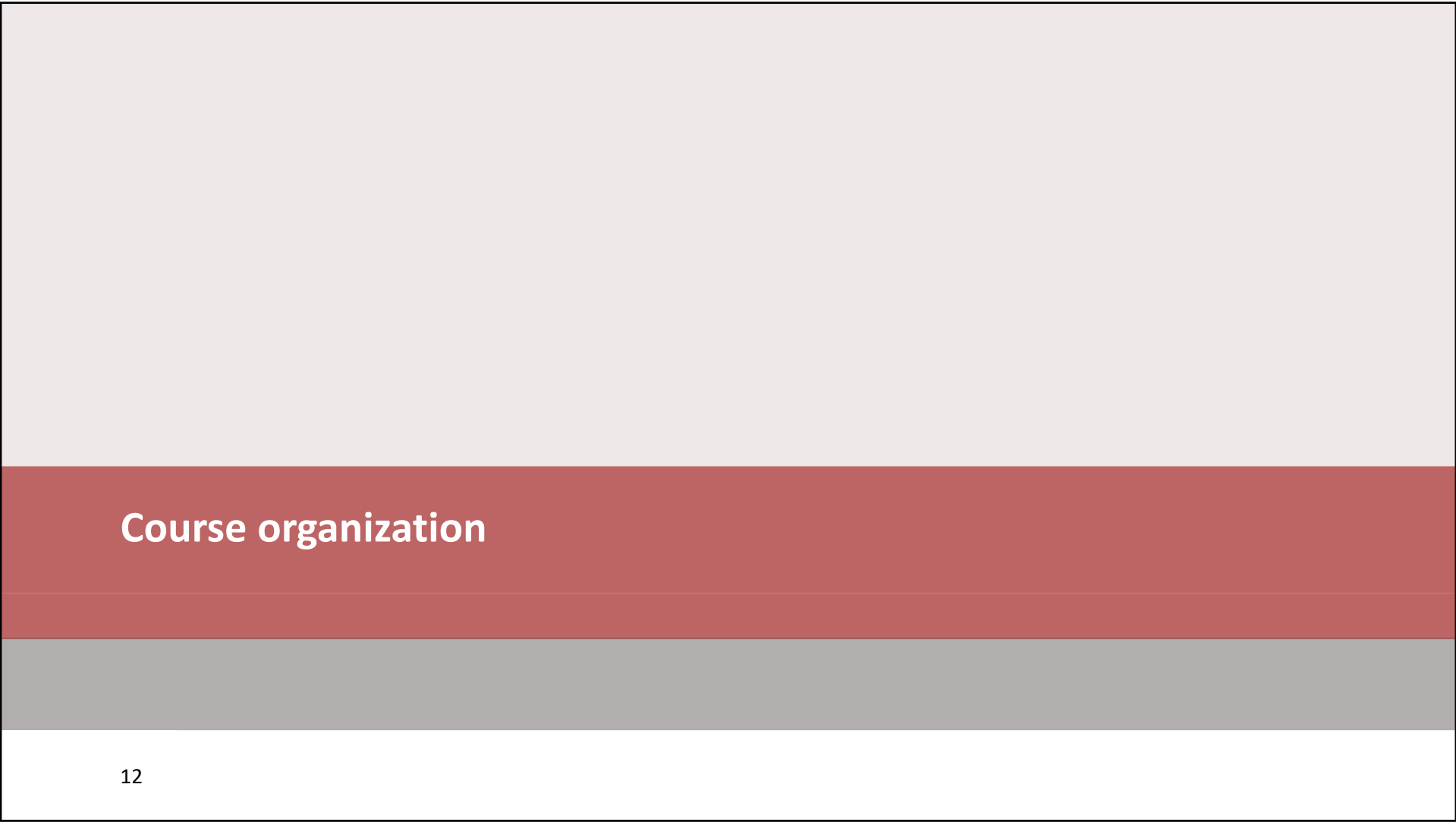
In this course:

We discuss various approaches to

- **collect,**
- **process,**
- **store, and**
- **analyze**

data for creating AI/ML based engineering systems and solutions.

Remember: data and knowledge do not come cheap!



Learning objectives

- **design a basic data acquisition system for goal-targeted data collection**
- **pre-process data for further analysis**
- **perform feature engineering**
- **apply basic data analysis techniques**
- **build machine learning models for clustering, classification and regression**
- **develop non-linear machine learning models in Python**
- **present the results of data analysis in writing**

Lecturers and teaching assistants

 Prof.dr.ir. U. Kaymak Course Coordinator; Lecturer u.kaymak@tue.nl	 Dr. F. van der Sommen, MSc. Lecturer fvdsommen@tue.nl	 Dr.ir. L.A. Zavala Mondragon Lecturer l.a.zavala.mondragon@tue.nl	 Q. Khaled Abo Shama, MSc. Teaching Assistant q.k.h.abo.shama@tue.nl
 P.C. Gort, MSc.	 L. Abdi, MSc.	 S.Chinniah	 G. Anesiadis

Meetings

15 sessions

- typically, 2 x 2 hrs./week lecture, 2 x 2 hrs./week instruction, 8 weeks long, except for the week of 21 September (see schedule for the latest information)

Tuesday 1 – 4, various rooms, Friday 5 – 8, various rooms

Lectures: introduce and explain main concepts

Instructions for (practice) exercises and working on the assignment

Q&A session at the end of the quartile

Further questions can be asked through Canvas (preferred method) or during instructions

Planning
(partial)

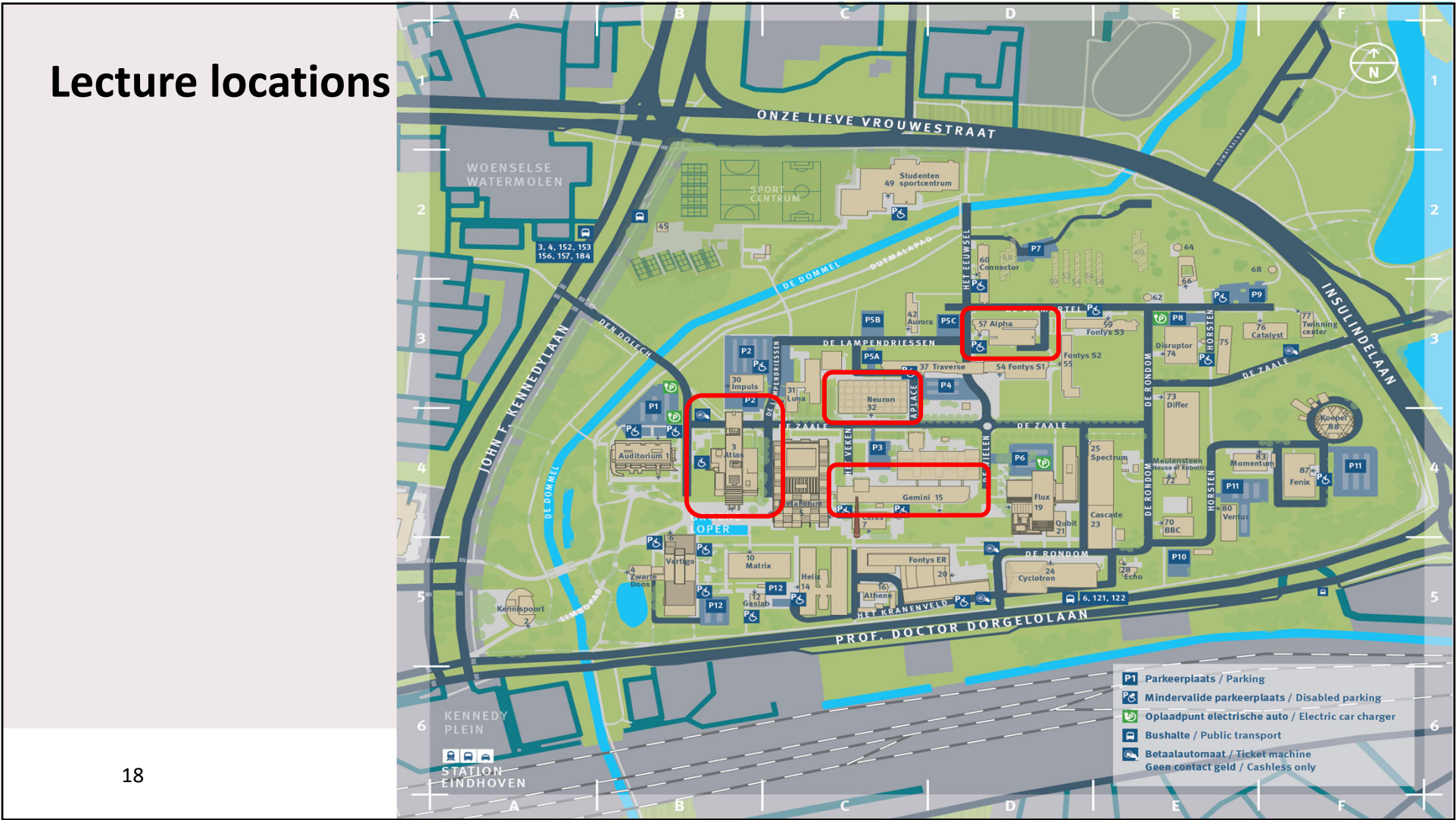
WEEK	DAY/TIME	ROOM	TYPE	TOPIC
1	01-09-2026 / 08:45 - 10:30	Alpha 0.98	Lecture	Introduction to the course, basic learning types, basic ML tasks
	01-09-2026 / 10:45 - 12:30	Atlas -1.715	Instruction	Team composition, software installation
	04-09-2026 / 13:30 - 15:15	Gemini-Noord 1.610	Lecture	Basic data collection, data representation, pre-processing
	04-09-2026 / 15:30 - 17:15	Gemini-Noord 1.610	Instruction	Various exercises, introduction to Assignment 1
...	...	...	...	...
	04-11-2026 / 13:30 - 16:30	To be announced	Exam	
	27-01-2027 / 18:00 - 21:00	To be announced	Resit	

ccessibility

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Always check CANVAS and OSIRIS for the latest information!

TU/e



Course material (literature and tools)

- Material provided by lecturers, e.g. slides, handouts, etc.
- Scientific papers
- Industry white papers
- Self-study (video) tutorials (recommended)
- Python exercises

Assessment

Components:

Assignment 1 – 20%

- Deadline: October 4th, 2026

Assignment 2 – 20%

- Deadline: October 25th, 2026

Written exam – 50%

Weekly quizzes – 10% (announcements through Canvas)

Assignments will be made in groups of 3.

Register through Canvas as soon as possible

It is not possible to re-sit assignments
Assignments are valid only in the
current academic year

**A minimum grade of 5.0 for the
written exam is needed to pass the course.**

Exam

Type: written, on paper, closed book

Date: 4 November 2026, **13.30 – 16.30**

Re-sit: 27 January 2027, **18.00 – 21.00**

Code of Conduct

Students are bound by TU/e Code of Conduct. When you submit your work under your own name you are asserting ownership of that work. When using ideas of another person, you must give that person appropriate credit through referencing. Referencing serves multiple purposes: (i) it allows readers to further explore sources you have consulted, (ii) it shows the depth of your own thinking and process of inquiry, (iii) it allows you and your readers to compare and contrast your position with other people's positions, agreeing with some, disagreeing with others, and (iv) it gives proper credit to the hard work that many people have done before you. Make sure you avoid any appearance of plagiarism in your work.

Policy on the use of AI tools

For the assignment, you may use tools such as ChatGPT to review completed drafts for editing, i.e.

- check narration or subtitles in videos/audio for grammar, clarity, and flow;
- review instructions, labels, or UI text in prototypes for clarity in the prototypes;
- check code comments, variable names, or textual descriptions in scripts, programming, or visualizations for readability and overall coherence.

It is not allowed to have AI generate complete texts, tables, or graphs instead of making them yourself. AI tools may only be used for supportive purposes.

- **Transparency:** document clearly for each tool what you used it for in the technology statement. **You must use the Technology statement student(s) (see text below).** Incomplete transparency or presenting AI output as your own work is not permitted.
- **Data confidentiality:** never enter sensitive or confidential data into AI platforms, as these may not comply with the university's or external organizations' privacy guidelines.

Improper or non-transparent use of AI within the assignment will lead to sanctions and may result in your grade being declared invalid by the Examination Board.

Technology statement student(s)

For the assignment, you must add the technology statement, using the text below. Replace the text in capital letters with the requested information.

During the preparation of this work, I/We used [NAME TOOL / SERVICE / VERSION OF AI TOOL] in order to [REASON]. The following parts of the assignment were affected/generated by AI tool usage: [INTRODUCTION / METHODS / xxx, DISCUSSION]. After using this tool/service, [NAME STUDENT(S)] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [NAME STUDENT (S)] take(s) full responsibility for the content of their work.

Position in curriculum

Topic: data science

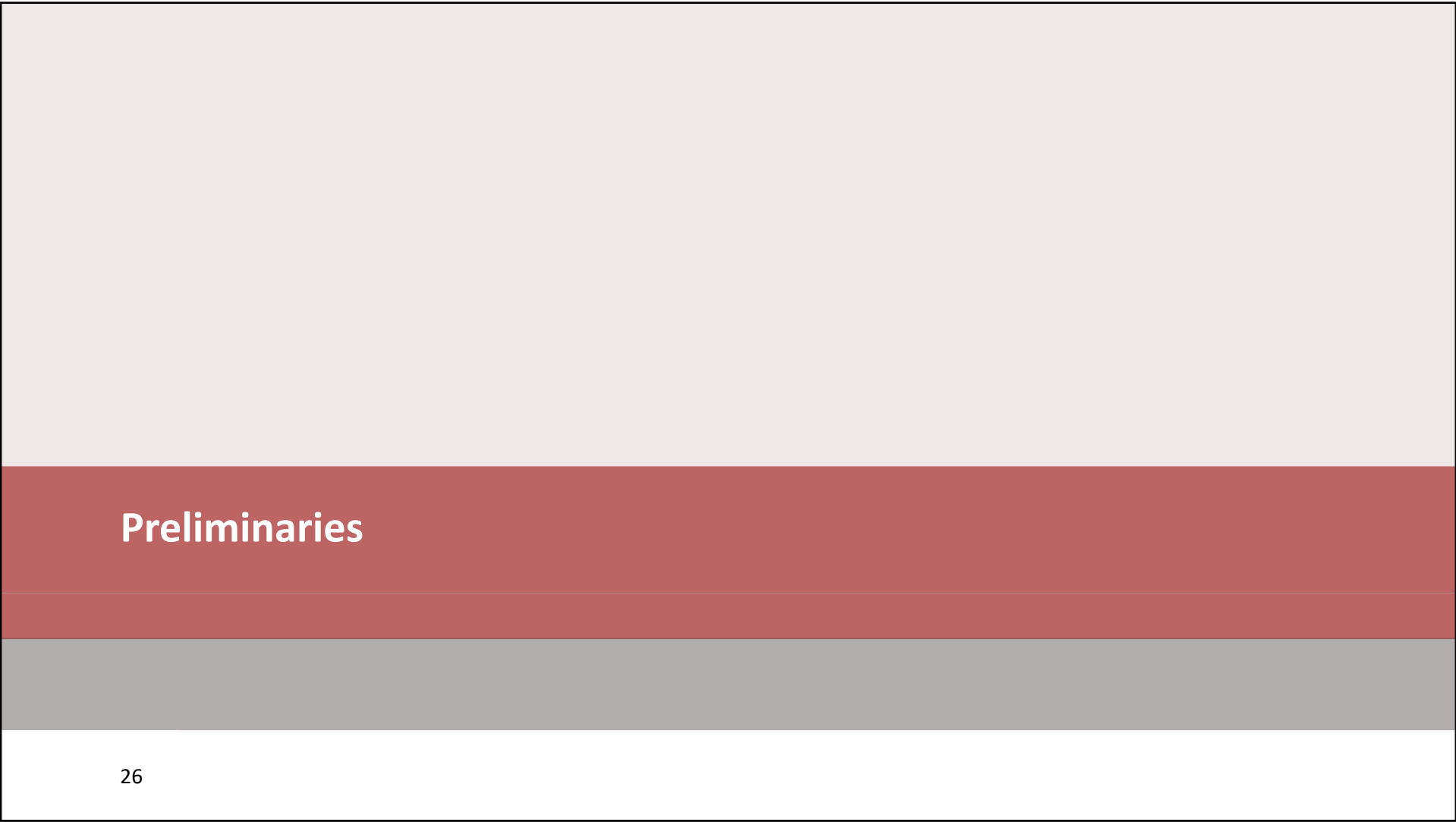
Course name: Data analysis and learning methods

Credits: 5 ECTS

Core course in Master program Artificial Intelligence & Engineering Systems

- **Year 1, Q1**

Input towards multiple courses (core and elective)



Two MSc programs

Data Science and Artificial Intelligence

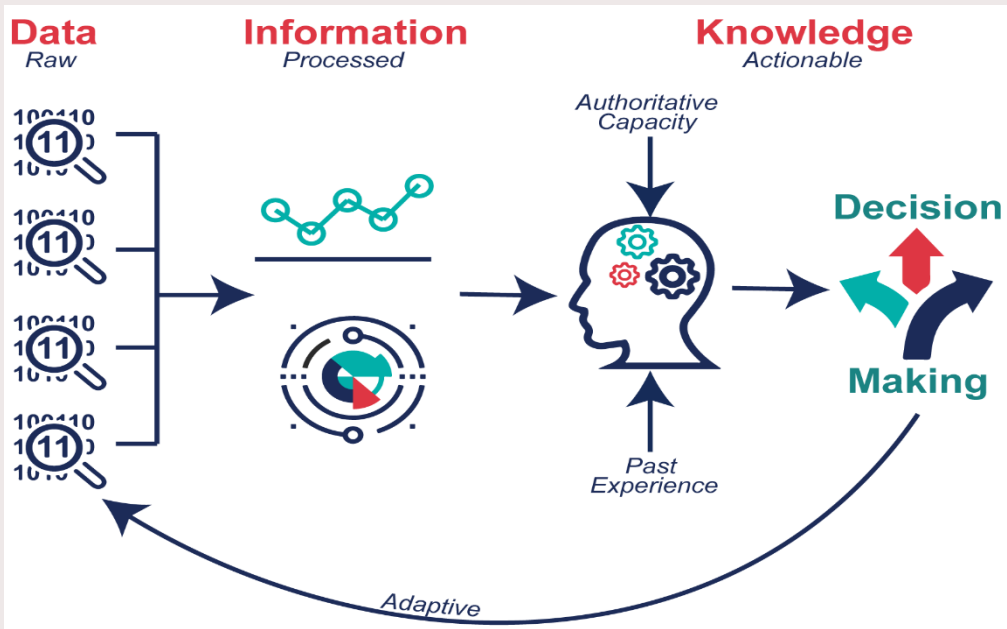
Artificial Intelligence and Engineering Systems

Types of data collection

Primary data collection – data collected for the first time, for a particular purpose

Secondary data collection – re-use of data that has already been collected for another purpose

From raw data to wisdom



Data collection influenced by:

- task
- perspective
- goal

Based on Ackoff, R. L., "From Data to Wisdom", Journal of Applied Systems Analysis, Volume 16, 1989 p 3-9.

Data

- What is a datum?
- When is something data?
- What do you need to make sense from the data?
- Components to record?

→ Data models

Data model



An abstract model that organizes elements of data and standardizes how they relate to one another and to the properties of real-world entities

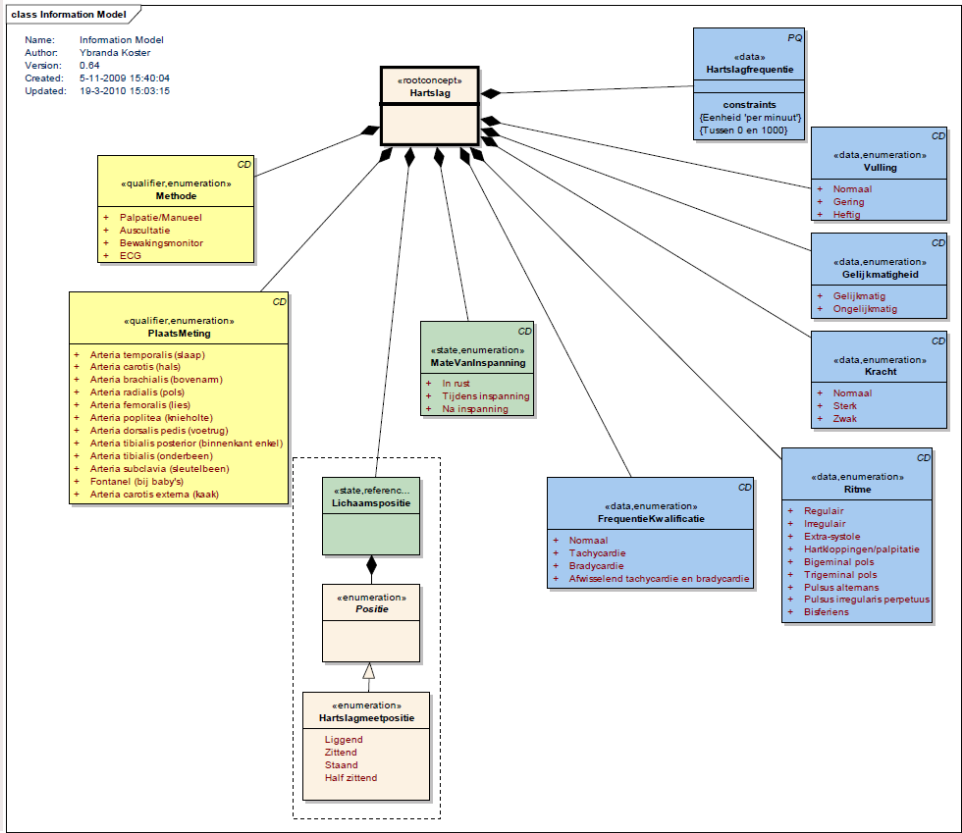


Aid communication between domain experts and technical experts



Show the data needed and created by business processes

Example: data model for measuring heart rate



- Information elements:
- Heart rate frequency
 - Heart condition qualification
 - Heart rhythm
 - Method of measurement
 - Place of measurement
 - Body posture
 - ...

Aspects of data collection



Illustration: iStock



Royal Geographical Society

Machine learning basics

Some slides by G. Manias and D. Di Nucci

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What is machine learning?

Oxford definition:

the use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data

Solved through an optimization problem

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Types of learning

Basic learning types:

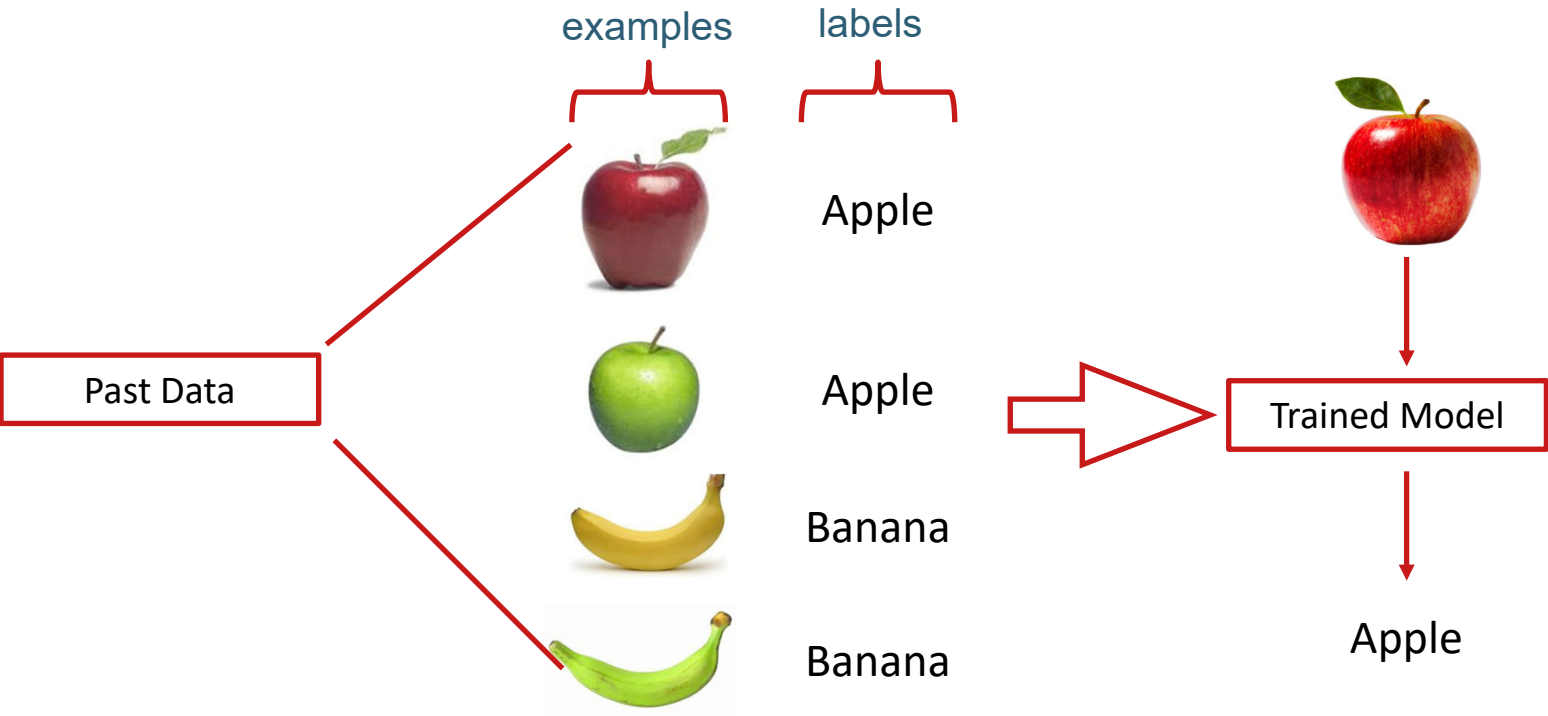
- Supervised
- Unsupervised
- Reinforcement learning

Other learning types:

- Semi-supervised learning
- Transfer learning
- Active learning
- Etc.

[38]

An Example of Supervised Learning

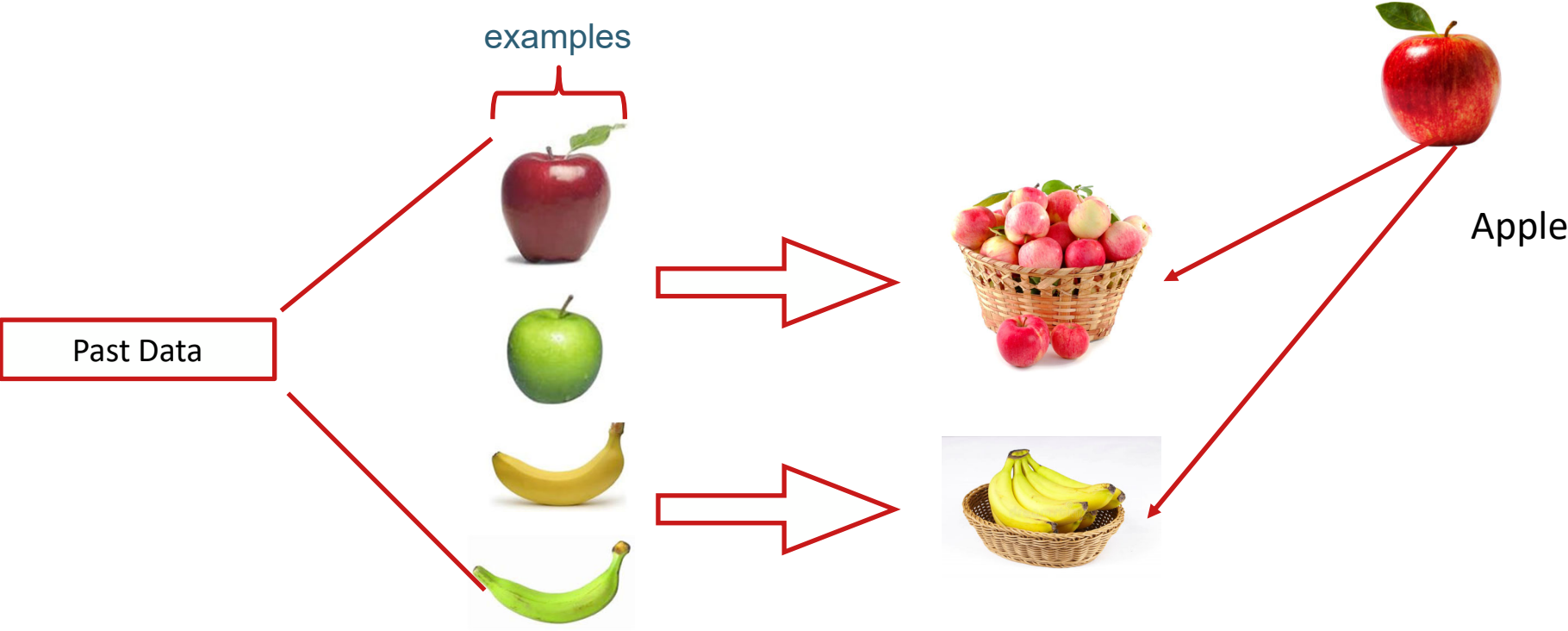


Given a labeled dataset, the model learns to predict new examples.

[Slide courtesy of Prof. Dario Di Nucci](#)



An Example of Unsupervised Learning

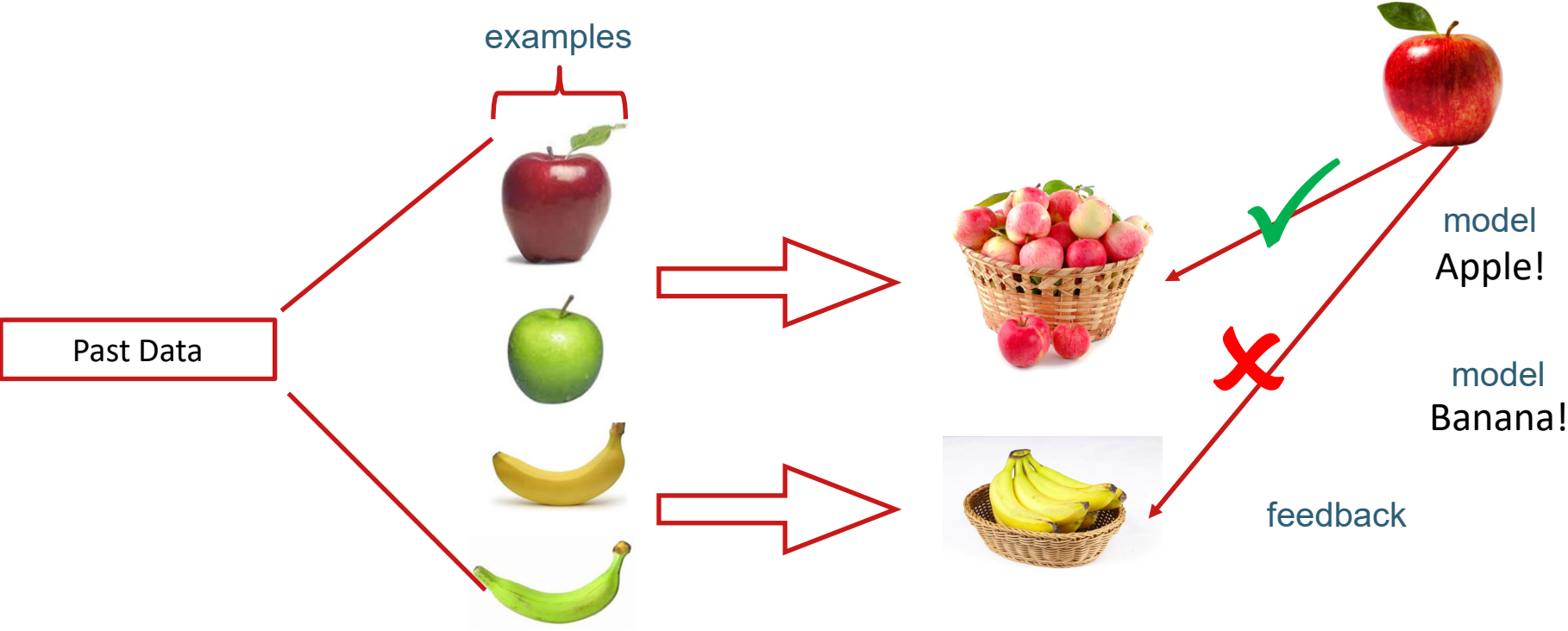


Given a dataset, without labels, the model learns to cluster/group similar data.

[Slide courtesy of Prof. Dario Di Nucci](#)



An Example of Reinforcement Learning



Given a dataset, the model learns to take correct actions (predictions) from feedback.

Machine learning tasks

Classification

Regression

Similarity matching

Clustering

Anomaly detection

Co-occurrence grouping

Profiling

Link prediction

Data reduction

Causal modeling

[42]

Common learning methods (models)

Linear regression

Logistic regression

Decision trees

k nearest neighbors classifier

Naïve Bayes classifier

(Nonlinear) input – output mappings

Mixture models

Support vector machines

Neural networks

Fuzzy inference systems

Bayesian networks

[43]

Recap

- **Data are nowadays ubiquitous and have multiple facets**
- **Distinction between primary data collection and secondary data collection**
 - In engineering systems, primary data collection has focus
 - Data science deals also with secondary data collection
- **Learning is related to solving an optimization problem**
- **Three main types of learning**
- **Many different machine learning tasks**